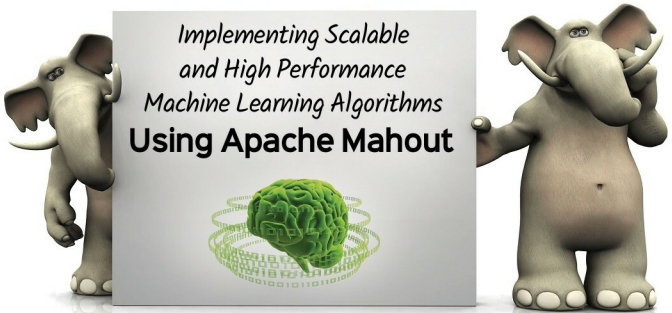




Apache Mahout aims at building an environment for quickly creating scalable and performant machine learning applications.

Machine learning refers to the intelligent and dynamic response of software or embedded hardware programs to input data. Machine learning is the specialised domain that operates in association with artificial intelligence to make strong predictions and analyses. Using this approach, there is no need to explicitly program computers for specific applications; rather, the computing modules evaluate the data set with their inherent reactions so that real-time fuzzy based analysis can be done. The programs developed with machine learning paradigms focus on the dynamic input and data set, so that the custom and related output can be presented to the end user.

There are a number of applications for which machine learning approaches are widely used. These include fingerprint analysis, multi-dimensional biometric evaluation, image forensics, pattern recognition, criminal investigation, bioinformatics, biomedical informatics, computer vision, customer relationship management, data mining, email filtering, natural language processing, automatic summarisation, and automatic taxonomy construction. Machine learning also applies to robotics, dialogue systems, grammar checkers, language recognition, handwriting recognition, optical character recognition, speech recognition, machine translation, question answering, speech synthesis,

text simplification, pattern recognition, facial recognition systems, image recognition, search engine analytics, recommendation systems, etc.

There are a number of approaches to machine learning, though traditionally, supervised and unsupervised learning are the models widely used. In supervised learning, the program is trained with a specific type of data set with the target value. After learning and deep evaluation of the input data and the corresponding target, the machine starts making predictions. The common examples of supervised learning algorithms include artificial neural networks, support vector machines and classifiers. In the case of unsupervised learning, the target is not assigned with the input data. In this approach, dynamic evaluation of data is done with high performance algorithms, including k-means, self-organising maps (SOM) and clustering techniques. Other prominent approaches and algorithms associated with machine learning include dimensionality reduction, the decision tree algorithm, ensemble learning, the regularisation algorithm, supervised learning, artificial neural networks, and deep learning. Besides these, there are also the instance-based algorithms, regression analyses, classifiers, Bayesian statistics, linear classifiers, unsupervised learning, association rule learning, hierarchical clustering, deep